

Employee Attrition Analysis – Key Findings

Project Overview

The objective of this project was to analyze employee data and identify the factors contributing to employee attrition. The analysis aimed to understand workforce turnover patterns and develop machine learning models capable of predicting whether an employee is likely to leave the organization.

Data Preparation

- Loaded and explored the IBM HR Attrition dataset.
- Checked for missing values and confirmed that the dataset was complete.
- Identified numerical and categorical variables.
- Removed irrelevant features.
- Encoded categorical variables.
- Split the dataset into 80% training and 20% testing data.

Exploratory Data Analysis

- Most employees remained with the company while a smaller percentage left.
- Attrition varied across departments and job roles.
- Lower-income employees were more likely to leave.
- Work-life balance and years at the company influenced attrition.

Important Factors Affecting Employee Attrition

1. Monthly Income
2. Work-Life Balance
3. Years at Company
4. Job Role
5. Department
6. Overtime

Model Performance

Models used: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier. Evaluation metrics included Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Key Insights

- Attrition is influenced by multiple factors.
- Poor work-life balance increases turnover.
- Newer employees are more likely to leave.
- Predictive models can identify at-risk employees.

Recommendation

Organizations should improve employee engagement, work-life balance, and career development opportunities while monitoring key attrition indicators.

Conclusion

This analysis demonstrated how data analytics and machine learning can be used to predict employee attrition and support data-driven HR decision-making.